

Changes in flow-mediated brachial artery vasoactivity with lowering of desirable cholesterol levels in healthy middle-aged men.

Current National Cholesterol Education Program guidelines consider desirable total and low-density lipoprotein cholesterol levels to be < 200 and < 160 mg/dl, respectively, for healthy individuals without multiple coronary risk factors. To determine the extent to which these levels affect vascular function, we assessed flow-mediated (endothelium-dependent) brachial artery vasoactivity noninvasively before, during, and after cholesterol lowering (simvastatin 10 mg/day) in 7 healthy middle-aged men with cholesterol levels meeting current recommendations. Flow-mediated brachial artery vasoactivity was measured using 7.5 MHz ultrasound and expressed as percent diameter change from baseline to hyperemic conditions (1 minute following 5 minutes of blood pressure cuff arterial occlusion). Flow-mediated vasoactivity rose from 5.0 $\pm$ 3.6% at baseline to 10.5 $\pm$ 5.6%, 13.3 $\pm$ 4.3%, and 15.7 $\pm$ 4.9% (all p < 0.05) as cholesterol fell from 200 $\pm$ 12 to 161 $\pm$ 18, 169 $\pm$ 16, and 153 $\pm$ 11 mg/dl after 2, 4, and 12 weeks, respectively, of cholesterol-lowering therapy. Vasoactivity and cholesterol returned to baseline levels 12 weeks after simvastatin discontinuation. Overall, vasoactivity was found to correlate inversely with cholesterol levels (r = -0.47, p = 0.004). These data suggest that flow-mediated brachial artery vasoactivity responds rapidly to changes in cholesterol levels and that endothelial function improves by lowering cholesterol levels below recommendations of current guidelines.